Ch 8.1: Decision Trees

Lecture 23 - CMSE 381

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Announcements

Last time:

- Cubic Splines

This lecture:

- 8.1 Decision Trees

Announcements:

- HW #7 Due tonight
-

Section 1

Decision Trees

Big idea

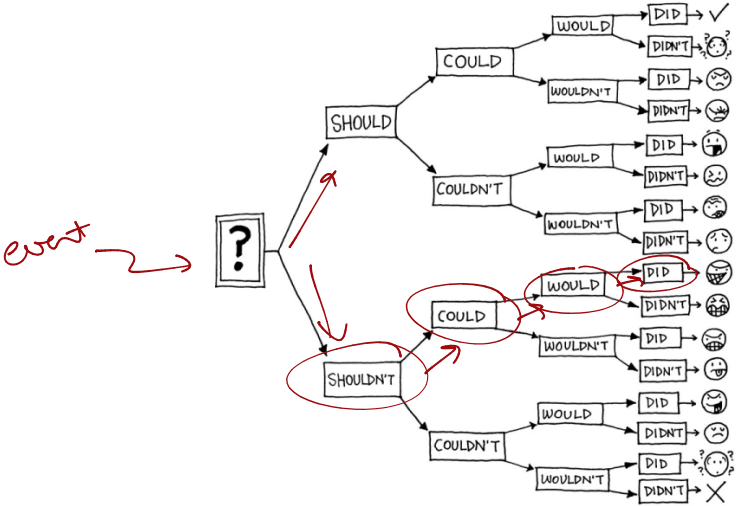

Image: https://marekbennett.com/2014/02/14/decision-tree/

Subset of Hitters data

	Hits	Years	Salary	LogSalary
1	81	14	475.0	6.163315
2	130	3	480.0	6.173786
3	141	11	500.0	6.214608
4	87	2	91.5	4.516339
5	169	11	750.0	6.620073
...	...	...	...	...
317	127	5	700.0	6.551080
318	136	12	875.0	6.774224
319	126	6	385.0	5.953243
320	144	8	960.0	6.866933
321	170	11	1000.0	6.907755

transforming by a log results something closer to normal dist.

Predict log(salary) = y

X_1 = Hits

X_2 = Years

report

e^y = salary

First decision tree example

	Hits	Years	LogSalary
1	81	14	6.163315
2	130	3	6.173786
3	141	11	6.214608
4	87	2	4.516339
5	169	11	6.620073
...	...	...	...
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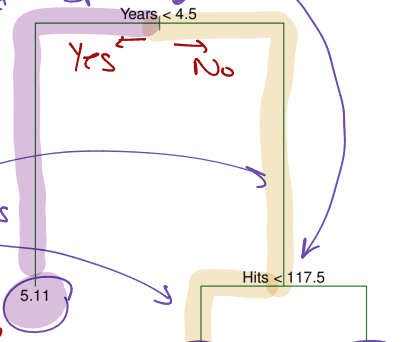
Is years < 4.5?
No, 14 > 4.5 right
Is Hits < 117.5?
Yes left

splits

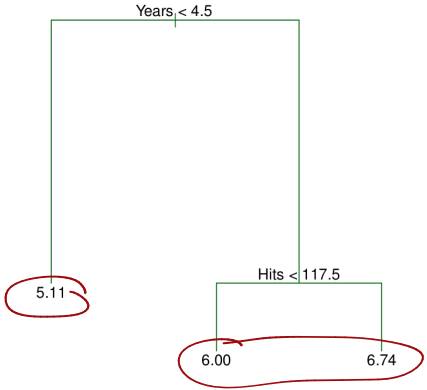


branches

log salary = 5.11
pred = $e^{5.11}$

leaves

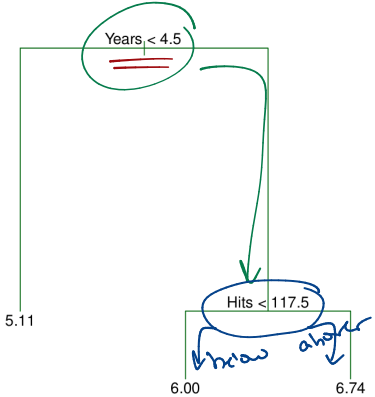
Interpretation of example

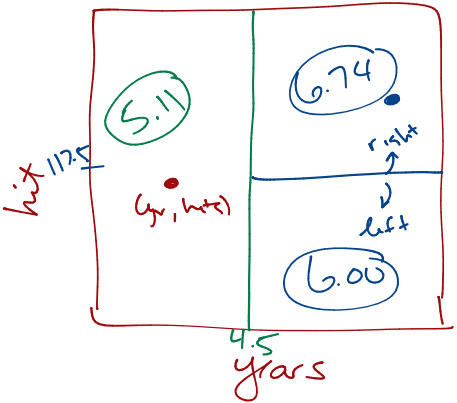


- First split says years is more important for predicting Salary
- Players with less experience lower Salary
- More experienced players are affected by # hits
- Very oversimplified!

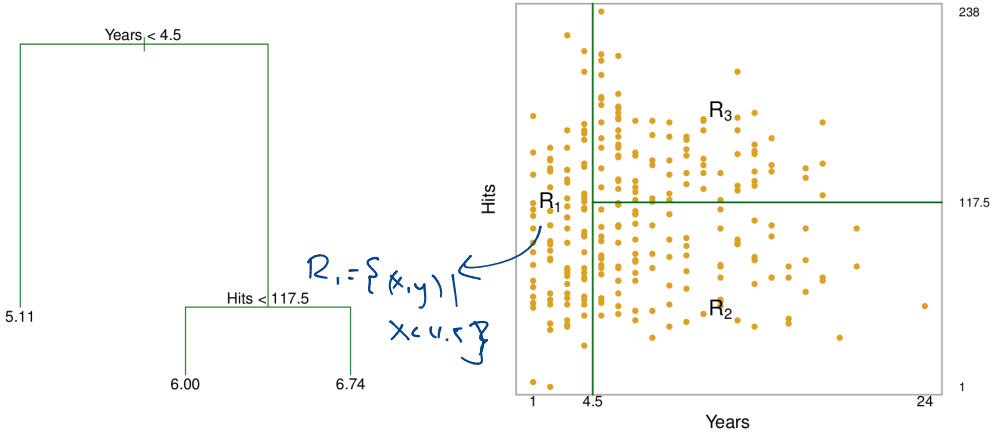
Coding a regression decision tree

Regions defined by the tree





Viewing Regions Defined by Tree

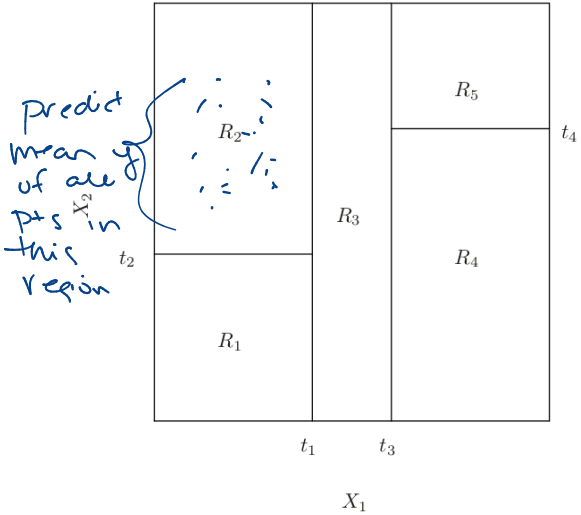


$R_1 = \{(x,y) \mid x < 4.5\}$

How do we actually get the tree? Two steps

1. We divide the predictor space – that is, the set of possible values for $X_1, X_2, \cdots, X_p$ — into J distinct and non-overlapping regions, $R_1, R_2, \cdots, R_J$.
2. For every observation that falls into the region R_j, we make the same prediction = the mean of the response values for the training observations in R_j.

predict mean of all of pts in this region



Step 1: How do we decide on R_js?

Can't check every possible partition of the space

Goal:

Find boxes $R_1, \cdots, R_J$ that minimize

Greedy algorithm: Split the space in half using the "best" split each time

$$\sum_{j=1}^{J} \sum_{i \in R_j} (y_i - \hat{y}_{R_j})^2$$

i = data pts who is in set R_j

Repeat...

$\hat{y}_{R_j}$ = mean response for training observations in jth box

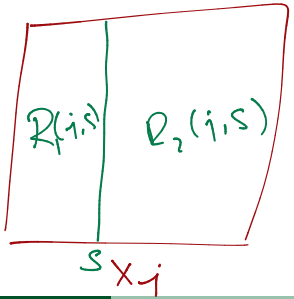
This is MSE!

Recursive Binary Splitting

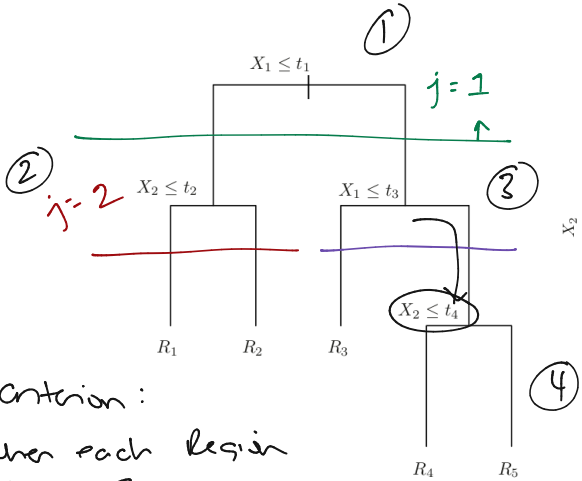
- Pick X_j one of my input variables
- Pick s so that splitting into $\{X \mid X_j < s\}$ and $\{X \mid X_j \geq s\}$ results in largest possible reduction in RSS

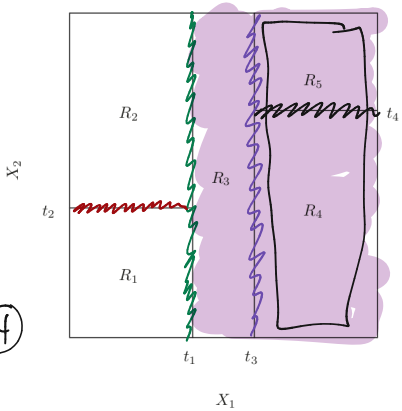
$$R_1(j,s) = \{X \mid X_j < s\}$$
$$R_2(j,s) = \{X \mid X_j \geq s\}$$

$$RSS = \sum_{i|x_i \in R_1(j,s)} (y_i - \hat{y}_{R_1})^2 + \sum_{i|x_i \in R_2(j,s)} (y_i - \hat{y}_{R_2})^2$$



Rinse and repeat



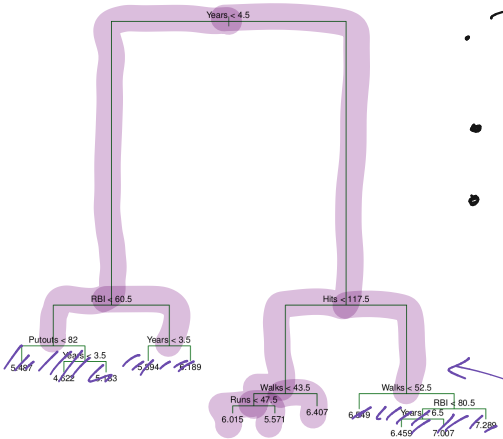


Stopping criterion:

e.g. Stop when each Region only has 5 pts

e.g. Stop when split 3 times..

Pruning



- Big trees may potential for overfitting
- Stop building at a depth
- Instead, could grow a big tree, prune it back based on improving the test error rate

Subtree is a subset of the tree

Weakest Link Pruning

Also called Cost complexity pruning

For every α, there is a subtree T that minimizes:

RSS

$$\sum_{m=1}^{|T|} \sum_{i|x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$$

- $|T|$ = number of terminal nodes of T
- R_m is rectangle for mth terminal node
- $\hat{y}_{R_m}$ is mean of training observations in R_m

Algorithm version

Algorithm 8.1 *Building a Regression Tree*

1. Use recursive binary splitting to grow a large tree on the training data, stopping only when each terminal node has fewer than some minimum number of observations.

2. Apply cost complexity pruning to the large tree in order to obtain a sequence of best subtrees, as a function of α.

3. Use K-fold cross-validation to choose α. That is, divide the training observations into K folds. For each $k = 1, \ldots, K$:

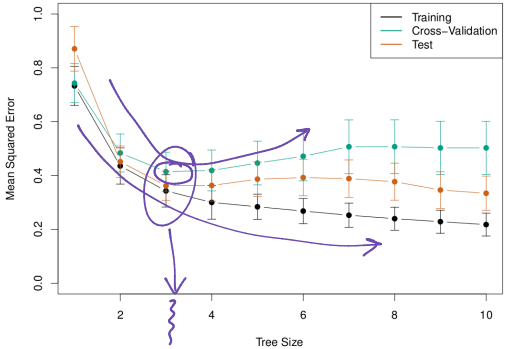
   (a) Repeat Steps 1 and 2 on all but the kth fold of the training data.

   (b) Evaluate the mean squared prediction error on the data in the left-out kth fold, as a function of α.

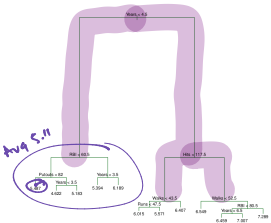
   Average the results for each value of α, and pick α to minimize the average error.

4. Return the subtree from Step 2 that corresponds to the chosen value of α.

Messing with α



3 nodes .... leaves

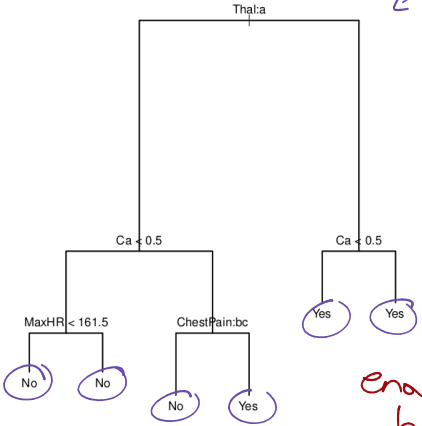


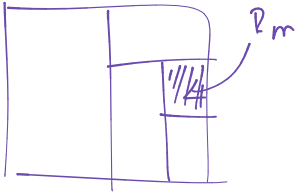
Section 2

Classification Decision Tree

Basic idea

heart data set





- $\hat{p}_{mk}$ = proportion of training observations in R_m from the kth class
- $E = 1 - \max_k(\hat{p}_{mk})$

fraction of the training obs not in the most common class in R_m

Not a sensitive enough measure for building the tree

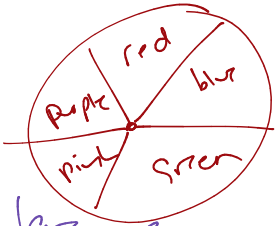
Gini index

Class labels to be predicted

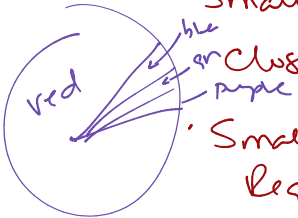
$$G = \sum_{k=1}^{K} \hat{p}_{mk}(1 - \hat{p}_{mk})$$

Proportion pts in R_m labeled k

- Region R_m
- Measuring total variance accross the K classes
- Small value if $\hat{p}_{mk}$ is close to 0 or 1
- Small value means Region contains obs from mostly one class



large G



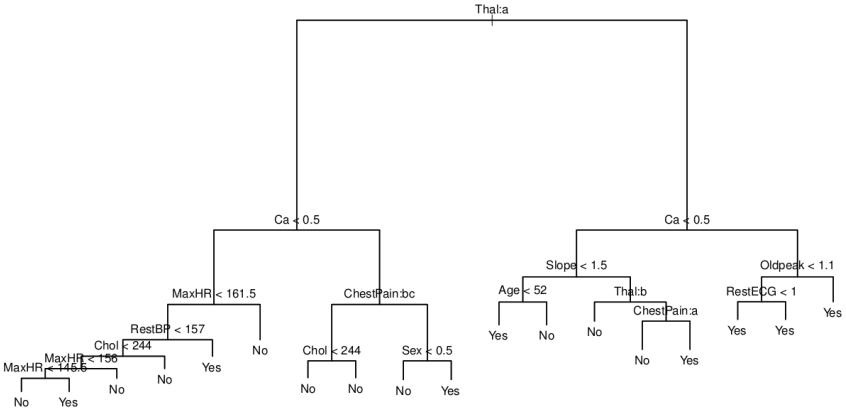
Small G

Entropy

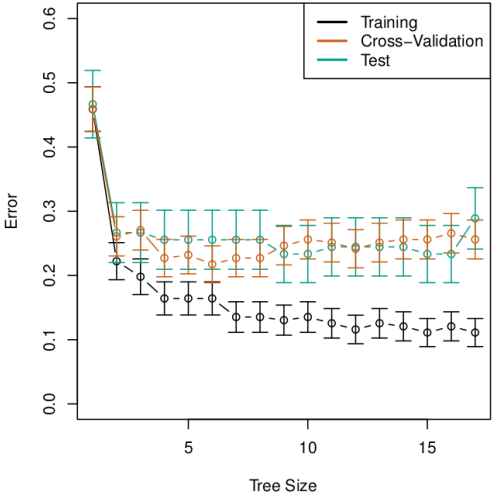
$$D = -\sum_{k=1}^{K} \hat{p}_{mk} \log \hat{p}_{mk}$$

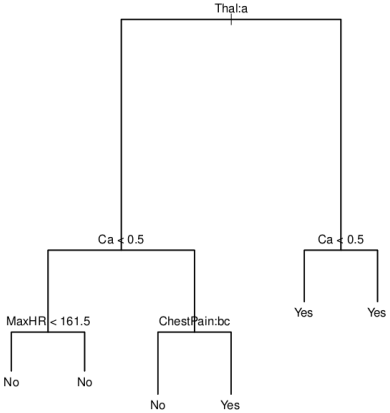
Another option

Example


Thal:a
Ca < 0.5
Ca < 0.5
MaxHR < 161.5
ChestPain:bc
RestBP < 157
Chol < 244
MaxHR < 156
MaxHR < 145.5
No
Yes
No
No
No
Yes
Chol < 244
Sex < 0.5
No
No
No
Yes
Slope < 1.5
Oldpeak < 1.1
Age < 52
Thal:b
RestECG < 1
Yes
Yes
Yes
ChestPain:a
Yes
No
No
No
Yes

Pruning the example

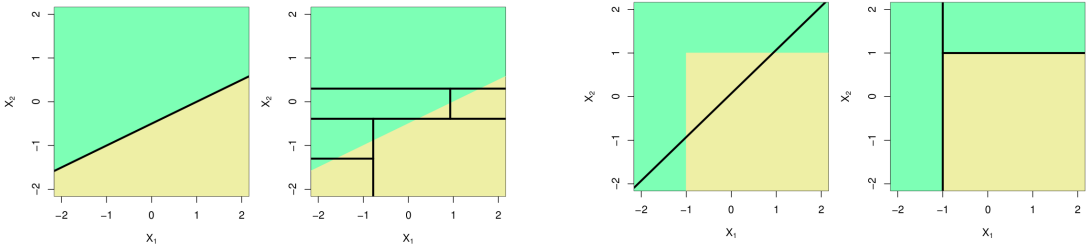




More coding!

Linear models vs trees

Start here next time



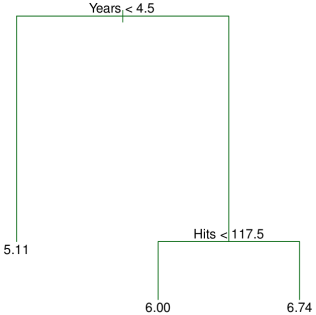
Pros/Cons

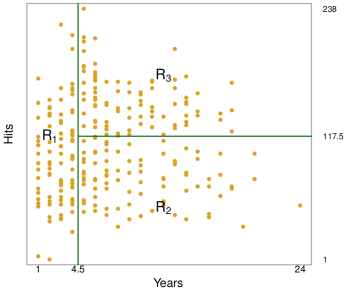
Pros:

Cons:

TL:DR

- Split into regions by greedily decreasing RSS
- Prune tree by using cost complexity
- Not robust - Next time, figure out how to aggregate trees





Next time

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Ensemble methods	8.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, 9.5	
28	W	Nov 23	Single layer NN	10.1	
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Multi Layer NN	10.2	HW #9 Due
30	W	Nov 30	CNN	10.3	
31	F	Dec 2	Unsupervised Learning & Clustering	12.1, 12.4	
32	M	Dec 5	More Clustering	12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	